Coordinate-Aware Point Cloud Registration for Crown-to-Root Alignment in CBCT and Intraoral Scans

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Abstract. Registering intraoral scans (IOSs) with cone-beam computed tomography (CBCT) links crown surfaces to root and bone anatomy, but the cross-modal point sets differ in coverage, density, and coordinate convention. We present an automatic rigid registration framework with deterministic point construction, separate hierarchical encoders, soft cross-modal correspondence, and differentiable rigid estimation. Public STSR 2025 Task 2 PointNet++–SVD weights initialize fine-tuning and the frozen pseudo-label teacher. Transformation-consistent augmentation jointly updates the source and target transformation, while entropy-based correspondence weighting preserves pretrained behavior. Official evaluation yielded 107.9250 mm mean translation error and 25.1658° mean rotation error. Controlled ablation favored supervised fine-tuning; pseudo-label quality remained a constraint, augmentation narrowed the semi-supervised gap, and entropy weighting preserved near-uniform pretrained behavior without a consistent advantage. Qualitative analysis linked large matrix translation errors to source-center sensitivity. The resulting pipeline identifies pseudo-label reliability and coordinate robustness as priorities.

Keywords: Cone-beam computed tomography · Intraoral scanning · Point cloud registration · Rigid transformation · Semi-supervised learning

1 Introduction

Cone-beam computed tomography (CBCT) depicts dental roots and surrounding bone, whereas an intraoral scan (IOS) captures crown surfaces at high detail. Registering them supports integrated tooth models and digital planning, motivating automatic fusion systems and clinical validation [3,7,9,13].

The modalities are not two clean observations of one surface: IOSs contain crowns, while thresholded CBCT points also include restorations and other dense anatomy. Their coverage, density, and coordinate origins differ. Classical closest-point and robust methods [2,4] therefore need reliable initialization or correspondences. Learned point features and matching [14,21] offer a direct

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alternative but remain sensitive to limited labels, partial overlap, and coordinate conventions.

STSR 2026 Task 2 provides labeled and unlabeled CBCT–IOS pairs. Although teacher predictions and confidence filtering can exploit unlabeled data [16,18], errors in a pseudo-transformation affect every supervised point. It is therefore necessary to test when pseudo-supervision improves rigid registration.

We present a pretraining-compatible pipeline with deterministic point construction, coordinate-handedness handling, hierarchical soft correspondence, and closed-form rigid estimation. We further introduce transformation-consistent augmentation and evaluate confidence-screened pseudo-labels and entropy weighting without altering the principal pretrained layers. Official evaluation, controlled ablations, resource measurements, and error analysis show that augmentation benefits semi-supervised configurations, while pseudo-label quality and entropy weighting remain configuration-sensitive relative to supervised fine-tuning.

2 Method

For each jaw, our method estimates one rigid transformation from an IOS source to a CBCT target. As summarized in Fig. 1, the shared prediction path constructs normalized point sets, extracts modality-specific descriptors, forms soft correspondences, and recovers a proper normalized rigid transformation by confidence-weighted SVD. The training-only branch combines labeled transformations with screened teacher predictions, applies transformation-consistent source augmentation, and fine-tunes the registration pipeline with a composite objective. At inference, the predicted transformation is restored to millimeter coordinates and validated before output.

2.1 Problem Formulation and Coordinate-Aware Point Construction

Let $X = \{x_i\}_{i=1}^{N_s} \subset \mathbb{R}^3$ be the IOS source and $Y = \{y_j\}_{j=1}^{N_t} \subset \mathbb{R}^3$ its CBCT target. The network estimates the normalized transformation

$$T_n = \begin{bmatrix} R & t_n \\ 0 & 1 \end{bmatrix}, \qquad R \in \mathrm{SO}(3), \quad t_n \in \mathbb{R}^3, \tag{1}$$

Here R is a proper 3D rotation, t_n is normalized translation, and the bottom row is homogeneous. The method outputs the millimeter-space matrix $T = [R, t; 0, 1]$, such that $Rx_i + t$ is expressed in CBCT coordinates. We deterministically sample $N_s = 4096$ IOS mesh vertices. CBCT voxels above intensity 600 are mapped through the image affine, then sampled to $N_t = 8192$ physical points.

We define the source center c_s as the midpoint of its axis-aligned bounding box, the target center c_t analogously, and the target scale $s = \max_j \|y_j - c_t\|_2$. Both modalities are normalized using this target scale, producing the two point sets shown in Fig. 1:

$$X^n = \{x_i^n\}, \quad Y^n = \{y_j^n\}, \qquad x_i^n = \frac{x_i - c_s}{s}, \quad y_j^n = \frac{y_j - c_t}{s}. \tag{2}$$

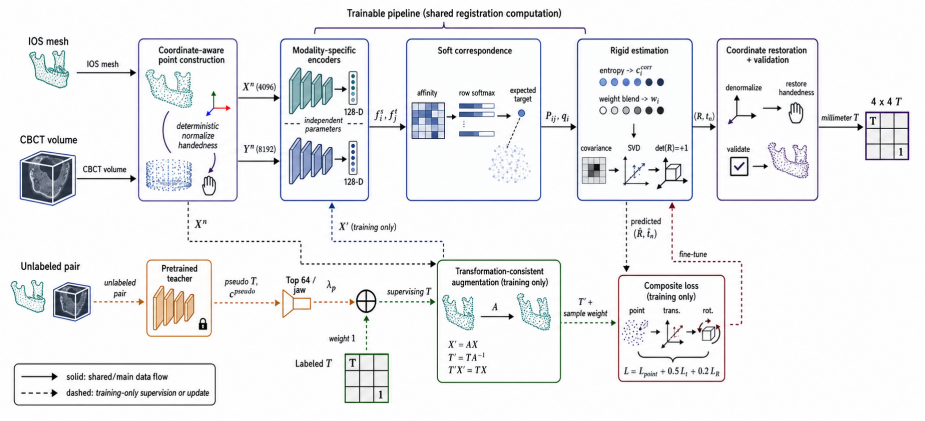


Fig. 1. Overview of the proposed registration framework. Solid arrows show the shared prediction and output path; dashed arrows show training-only supervision or updates. Separate encoders, soft correspondence, entropy confidence c_i^{corr}, and determinant-corrected SVD predict (R, t_n) from X^n and Y^n. Fine-tuning combines labeled and confidence-screened pseudo supervision (c^{pseudo}) with consistent augmentation $T'_n X' = T_n X^n$; the figure abbreviates T_n, T'_n as T, T'. Coordinate restoration yields the final millimeter T. The two jaws use independent networks.

The corresponding millimeter-space translation is

$$t = s\, t_n - R c_s + c_t. \tag{3}$$

Some labels encode a handedness change. When present, reflecting the first source coordinate by $F = \mathrm{diag}(-1, 1, 1)$ and right-multiplying the label by its homogeneous form lets the network estimate a proper rotation in SO(3). Training detects this condition from the label determinant. At inference, where labels are unavailable, a voxel-spacing pattern learned from the training set activates the same internal reflection; coordinate restoration later multiplies it back into the output. Both modes therefore map original IOS coordinates to CBCT coordinates, although this inference heuristic is acquisition dependent. The point-construction stage outputs only X^n, Y^n, and the metadata needed for restoration.

2.2 Modality-Specific Encoding and Soft Correspondence

Separate PointNet++ encoders [14] process the normalized source and target. They share an architecture but not parameters, permitting modality-specific features. Each progressively samples centers, aggregates increasing neighborhoods, and outputs 128-dimensional descriptors; at most 512 centers per modality enter matching. Upper and lower jaws use independent complete networks.

Let f_i^s and f_j^t denote source and target descriptors at encoded centers p_i and z_j, respectively. A learned linear map W places the source descriptor in the

target feature space. The soft-correspondence block computes the affinity and row-normalized probability

$$a_{ij} = (W f_i^s)^\mathsf{T} f_j^t, \qquad P_{ij} = \frac{\exp(a_{ij})}{\sum_k \exp(a_{ik})}. \tag{4}$$

Here a_{ij} and P_{ij} denote affinity and row-normalized probability; i indexes source centers, j and k index target centers, and $(\cdot)^\mathsf{T}$ denotes transpose. The expected target for source center p_i is

$$q_i = \sum_j P_{ij} z_j. \tag{5}$$

Thus the block outputs P_{ij} and q_i to rigid estimation. This follows learned soft correspondence in point-cloud registration [21]. The encoding and matching operations are identical during training and inference; only the training source may first be replaced by its augmented counterpart X'.

2.3 Confidence-Aware Rigid Estimation

Correspondence ambiguity is summarized by the normalized entropy

$$H_i = -\frac{\sum_j P_{ij} \log(P_{ij} + \epsilon)}{\log N_q}, \qquad c_i^{\text{corr}} = \max(1 - H_i, 0.05), \tag{6}$$

where N_q is the number of target centers, c_i^{corr} is correspondence confidence, and ϵ is a numerical constant. This symbol is distinct from the teacher confidence c^{pseudo} introduced below. For N_p source centers, let $w_i^u = 1/N_p$ and $w_i^{\text{corr}} = c_i^{\text{corr}} / \sum_k c_k^{\text{corr}}$. Their mixture is

$$w_i = (1 - \alpha) w_i^u + \alpha w_i^{\text{corr}}, \qquad \alpha = \sigma(\beta), \tag{7}$$

where σ is logistic and the learnable scalar β is initialized so that $\alpha \approx 0$, preserving the pretrained estimator's nearly uniform behavior.

Using w_i, we form weighted centroids $\bar{p} = \sum_i w_i p_i$ and $\bar{q} = \sum_i w_i q_i$, and cross-covariance

$$C = \sum_i w_i (p_i - \bar{p})(q_i - \bar{q})^\mathsf{T}. \tag{8}$$

For $C = U \Sigma V^\mathsf{T}$, $R = V D U^\mathsf{T}$ with $D = \text{diag}(1, 1, \det(V U^\mathsf{T}))$ enforces $\det(R) = +1$, and $t_n = \bar{q} - R\bar{p}$. This is least-squares rigid alignment [1,8] with differentiable matrix operations [6]. The block outputs (R, t_n) in both modes. During training this prediction is compared with supervision; during inference it proceeds directly to coordinate restoration.

2.4 Semi-Supervised Supervision and Consistent Augmentation

A frozen pretrained teacher predicts a normalized transformation T_n^{pseudo} and aggregate confidence c^{pseudo} for each usable unlabeled pair. Per jaw, the 64 most confident of 288 predictions are retained. A retained pseudo-labeled sample receives weight

$$\lambda_{\text{p}} = 0.15 + 0.35 \min(1, c^{\text{pseudo}}), \tag{9}$$

where λ_{p} denotes the pseudo-labeled sample weight, whereas a labeled transformation T_n has weight one. Merging these sources yields a supervising normalized transformation and a per-sample loss weight. This follows confidence-screened teacher training [16,18] and limits uncertain pseudo-labels. The teacher and confidence gate are used only to prepare training supervision.

During training only, a random rigid perturbation A is applied to the normalized IOS source. For either labeled or pseudo-labeled supervision, the source and its supervising transformation are updated together as

$$X' = AX^n, \qquad T_n' = T_n A^{-1}, \qquad T_n' X' = T_n X^n. \tag{10}$$

Here homogeneous point notation is used, X' is the augmented source, T_n' is its adjusted supervising transformation, and T_n denotes the selected labeled or pseudo transformation before augmentation. Each rotation axis is perturbed within $[-6^\circ, 6^\circ]$ and each normalized translation component within $[-0.06, 0.06]$. Equation (10) changes the source pose without changing its target-space correspondence. The augmented X' enters the source encoder, while Y^n is unchanged. Setting $A = I$ covers an unaugmented training sample. Inference omits the entire lower branch and uses deterministic X^n and Y^n.

2.5 Composite Training Objective

Let x_k' be a point from the source X' actually presented to the network, and let $T_n' = [R', t_n'; 0, 1]$ be its corresponding supervision from Eq. (10). The rigid-estimation block predicts $\hat{T}_n = [\hat{R}, \hat{t}_n; 0, 1]$. The transformed-point loss is

$$\mathcal{L}_{\text{point}} = \frac{1}{3N_s} \sum_{k=1}^{N_s} \|\hat{R}x_k' + \hat{t}_n - (R'x_k' + t_n')\|_2^2. \tag{11}$$

Here k indexes the N_s source points and $\mathcal{L}_{\text{point}}$ denotes their mean squared transformed-point discrepancy. Direct translation and rotation penalties are

$$\mathcal{L}_t = \text{SmoothL1}(\hat{t}_n, t_n'), \tag{12}$$

$$\mathcal{L}_R = \arccos\left[\text{clip}\left(\frac{\text{tr}(\hat{R}(R')^{\mathsf{T}}) - 1}{2}, -1 + \epsilon, 1 - \epsilon\right)\right]. \tag{13}$$

Here SmoothL1, tr, and clip denote the smooth L_1 penalty, matrix trace, and interval clamp; ϵ is the safeguard defined above. The per-sample objective is

$$\mathcal{L} = \mathcal{L}_{\text{point}} + 0.5\mathcal{L}_t + 0.2\mathcal{L}_R. \tag{14}$$

Batch losses use weight one for labeled samples and λ_{p} for pseudo-labeled samples. The point term constrains the combined transformation, while the translation and rotation terms directly supervise its components. Gradients pass through soft correspondence and the differentiable rigid estimator to fine-tune the learnable registration pipeline; the teacher remains fixed.

2.6 Coordinate Restoration and Output Validation

For each discovered case and jaw, the shared prediction path produces $T_n = [R, t_n; 0, 1]$. Equation (3) restores its translation to millimeters, after which any internal handedness conversion is mapped back to the original IOS convention. The resulting $T = [R, t; 0, 1]$ is a double-precision 4×4 matrix. It is checked for shape, finite entries, and homogeneous last row; case coverage and archive structure are also verified. This stage validates but does not refine predictions.

3 Experiments

3.1 Dataset and Metrics

The challenge provides 30 labeled, 300 unlabeled, and 50 hidden-label validation cases with upper- and lower-jaw inputs. A local audit retained 288 usable unlabeled pairs per jaw and logged invalid meshes. No external training dataset was used; permitted public predecessor checkpoints served only as initialization. Translation error is the Euclidean distance between predicted and reference translations in millimeters, while rotation error is their geodesic angular difference in degrees. Codabench reports the mean case-jaw errors; lower is better.

3.2 Implementation and Evaluation Design

Upper and lower networks were fine-tuned independently for 40 epochs. Submission models used all 30 labeled and 64 pseudo-labeled cases per jaw; with no remaining holdout, checkpoints were selected by weighted training loss. Undefined holdout log fields are therefore expected, not performance estimates. Table 1 gives the principal settings.

Component analysis held out six labeled cases per jaw and trained on 24. Five variants shared the split and seed: public pretrained, supervised, semi-supervised without rigid augmentation, complete semi-supervised, and complete with uniform SVD. Checkpoints minimized the sum of holdout translation and rotation errors. Because selection and reporting use the same holdout, this is component diagnosis rather than an unbiased generalization estimate. Organizer-provided online scores remain separate.

Pretrained estimator and checkpoints. Initialization used the official STSR 2025 Task 2 dual-branch PointNet++–SVD release [17] at commit `bb10b433bf13`. We loaded its released `best_model.pth` files from experiments: `lower_jaw_registration_v1`, `upper_jaw_registration_v1`. Model settings follow

Table 1. Training, environment, and model settings. FLOPs include only operations counted by the profiler and are therefore a lower bound.

Training setup	
Epochs / batch / seed	40 / 2 / 20260717
Optimizer / learning rate	AdamW [10] / 8×10^{-5}
Weight decay / scheduler / clip	10^{-5} / None / 5.0
Loss weights	point 1; translation 0.5; rotation 0.2
Environment and model	
OS / CPU / RAM	Debian 12 / i7-13700KF / 62 GiB
GPU / driver	RTX 4090 / 535.247.01
Python / PyTorch / CUDA	3.10.20 / 2.5.1+cu121 / 12.1
Parameters/jaw	82,433
Counted FLOPs/jaw	7.195 GFLOPs
Sequential train time	1309.466 s
Best epoch (upper/lower)	36 / 40

Table 2. Official online evaluation. Both geometric errors are minimized.

Metric	**Value**
Mean translation error	107.925014 mm
Mean rotation error	25.165843°

`configs/inference_config.yaml`: 4096 IOS points, 8192 CBCT points, and 128-D descriptors. All released encoder and attention weights initialized and were fine-tuned in the jaw-specific students; the same checkpoints were frozen for pseudo-label generation. The new entropy/uniform mixing scalar was the sole missing key and was initialized to -10; no unexpected keys were accepted. Full checkpoint hashes are recorded in the run metadata.

4 Results and Discussion

4.1 Official Performance and Computational Characteristics

Table 2 reports the two organizer-provided geometric errors.

The artifact reported 14.0 s total and 0.07 s/pair without hardware or timing boundaries, so it is not comparable with the local end-to-end measurement: 176.297 s for 50 cases (3.526 s/case) with a sampled 691 MiB GPU peak. Table 1 reports parameters and profiled FLOPs; the latter are a lower bound because neighborhood search and SVD are incompletely counted.

Table 3. Fixed-holdout component analysis over 12 case-jaw pairs used for checkpoint selection and reporting.

Variant	Pseudo	Rigid aug.	SVD weighting	Trans. (mm)↓	Rot. (°)↓	Train (s)
Public pretrained	No	No	Learnable, untrained	170.982	49.261	0.0
Supervised fine-tuning	No	Yes	Learnable	**105.023**	**17.853**	101.8
Semi-supervised, no aug.	Yes	No	Learnable	143.978	25.905	287.6
Complete semi-supervised	Yes	Yes	Learnable	137.941	24.870	287.8
Complete, uniform SVD	Yes	Yes	Uniform	136.795	24.530	287.6

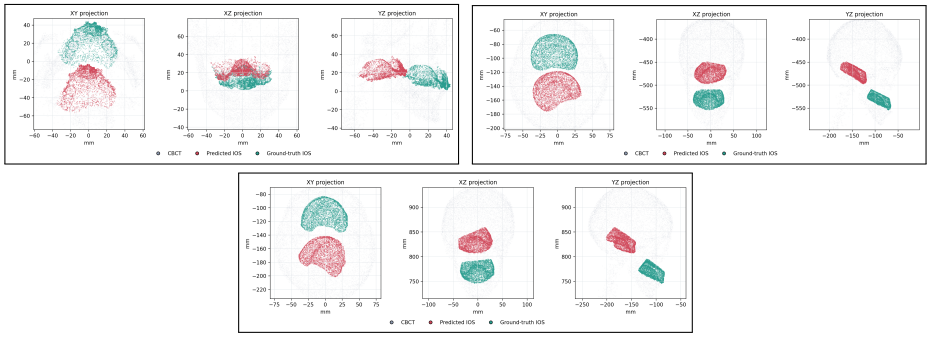


Fig. 2. Upper-jaw overlays: relative-good 269 (top left, 40.703 mm/20.117°), median 052 (top right, 69.147 mm/26.315°), and challenging 003 (bottom, 306.692 mm/25.237°). Gray is CBCT, red is predicted IOS, and green is ground truth; values are translation/rotation errors.

4.2 Component Ablation

On the fixed holdout (Table 3), supervised fine-tuning was best. The current pseudo-label configuration trailed it, while augmentation recovered part of the gap. Uniform SVD was stronger on this holdout; the entropy coefficient remained about 4.6×10^{-5}, indicating conservative preservation of near-uniform pretrained behavior whose adaptive benefit requires broader validation.

4.3 Qualitative Registration and Error Analysis

Figure 2 spans the six-case upper-jaw holdout; its labels indicate relative error levels rather than clinical categories.

Equation (3) shows how $-(\hat{R} - R)c_s$ amplifies rotation error for a distant IOS center: its norm was 804.5 mm for case 003 upper versus 58.7 mm for case 269. Case 003 therefore had 306.7 mm matrix translation error but 83.9 mm mean vertex displacement. This identifies one amplification mechanism, while positive rotation determinants point to factors beyond a simple reflection mismatch.

4.4 Limitations and Future Work

Current limitations include a single seed, reuse of six cases/jaw for selection and reporting, entropy ranking that needs stronger calibration, non-tooth CBCT

points, and acquisition-dependent handedness. Future work needs multi-seed evaluation, geometry-aware pseudo-labels, tooth extraction, explicit coordinate metadata, and robust low-overlap matching [5,15].

5 Conclusion

Our framework achieved 107.9250 mm translation and 25.1658° rotation error. Supervised fine-tuning was strongest; augmentation narrowed the pseudo-label gap, while entropy weighting remained close to its uniform initialization without a consistent advantage. Improving pseudo-labels and coordinate robustness remains a priority.

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References

1. Arun, K.S., Huang, T.S., Blostein, S.D.: Least-Squares Fitting of Two 3-D Point Sets. IEEE Transactions on Pattern Analysis and Machine Intelligence **PAMI-9**(5), 698–700 (1987). https://doi.org/10.1109/tpami.1987.4767965
2. Besl, P., McKay, N.D.: A method for registration of 3-D shapes. IEEE Transactions on Pattern Analysis and Machine Intelligence **14**(2), 239–256 (1992). https://doi.org/10.1109/34.121791
3. Elgarba, B.M., Fontenele, R.C., Ali, S., Swaity, A., Meeus, J., Shujaat, S., Jacobs, R.: Validation of a novel AI-based automated multimodal image registration of CBCT and intraoral scan aiding presurgical implant planning. Clinical Oral Implants Research **35**(11), 1506–1517 (2024). https://doi.org/10.1111/clr.14338
4. Fischler, M.A., Bolles, R.C.: Random sample consensus. Communications of the ACM **24**(6), 381–395 (1981). https://doi.org/10.1145/358669.358692
5. Huang, S., Gojcic, Z., Usvyatsov, M., Wieser, A., Schindler, K.: PREDATOR: Registration of 3D Point Clouds with Low Overlap. In: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4265–4274 (2021). https://doi.org/10.1109/cvpr46437.2021.00425
6. Ionescu, C., Vantzos, O., Sminchisescu, C.: Matrix Backpropagation for Deep Networks with Structured Layers. In: 2015 IEEE International Conference on Computer Vision (ICCV). pp. 2965–2973 (2015). https://doi.org/10.1109/iccv.2015.339
7. Jacobs, R., Salmon, B., Codari, M., Hassan, B., Bornstein, M.M.: Cone beam computed tomography in implant dentistry: recommendations for clinical use. BMC Oral Health **18**(1) (2018). https://doi.org/10.1186/s12903-018-0523-5
8. Kabsch, W.: A solution for the best rotation to relate two sets of vectors. Acta Crystallographica Section A **32**(5), 922–923 (1976). https://doi.org/10.1107/s0567739476001873

9. Liu, J., Hao, J., Lin, H., Pan, W., Yang, J., Feng, Y., Wang, G., Li, J., Jin, Z., Zhao, Z., Liu, Z.: Deep learning-enabled 3D multimodal fusion of cone-beam CT and intraoral mesh scans for clinically applicable tooth-bone reconstruction. Patterns **4**(9), 100825 (2023). https://doi.org/10.1016/j.patter.2023.100825
10. Loshchilov, I., Hutter, F.: Decoupled Weight Decay Regularization. arXiv preprint arXiv:1711.05101 (2017), https://arxiv.org/abs/1711.05101
11. Lumetti, L., Di Bartolomeo, M., Pellacani, A., Anesi, A., Grana, C., Bolelli, F.: Ontology-Grounded Structured Prediction for Dental CBCT Reporting. In: Medical Image Computing and Computer Assisted Intervention – MICCAI 2026 (2026)
12. Lumetti, L., Rizzo, F., Cremonini, F., Candeloro, E., Luca, L., Grana, C., Bolelli, F.: Do Multimodal LLMs Understand Intraoral Dental Data? Dataset, Platform, and Baselines. In: European Conference on Computer Vision (ECCV) (2026)
13. Preda, F., Nogueira-Reis, F., Stanciu, E.M., Smolders, A., Jacobs, R., Shaheen, E.: Validation of automated registration of intraoral scan onto Cone Beam Computed Tomography for an efficient digital dental workflow. Journal of Dentistry **149**, 105282 (2024). https://doi.org/10.1016/j.jdent.2024.105282
14. Qi, C.R., Yi, L., Su, H., Guibas, L.J.: PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space. arXiv preprint arXiv:1706.02413 (2017), https://arxiv.org/abs/1706.02413
15. Qin, Z., Yu, H., Wang, C., Guo, Y., Peng, Y., Xu, K.: Geometric Transformer for Fast and Robust Point Cloud Registration. In: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 11133–11142 (2022). https://doi.org/10.1109/cvpr52688.2022.01086
16. Sohn, K., Berthelot, D., Carlini, N., Zhang, Z., Zhang, H., Raffel, C.A., Cubuk, E.D., Kurakin, A.: FixMatch: Simplifying Semi-Supervised Learning with Consistency and Confidence. arXiv preprint arXiv:2001.07685 (2020), https://arxiv.org/abs/2001.07685
17. STSR Challenge Organizers: STSR Challenge: STSR 2025 Task 2 Baseline. https://github.com/ricoleehduu/STSR-Challenge (2025), directory STSR-2025/baseline/STSR-Task2-Baseline, commit bb10b433bf13
18. Tarvainen, A., Valpola, H.: Mean Teachers Are Better Role Models: Weight-Averaged Consistency Targets Improve Semi-Supervised Deep Learning Results. arXiv preprint arXiv:1703.01780 (2017), https://arxiv.org/abs/1703.01780
19. Wang, Y., Li, Z., Wu, C., Liu, J., Zhang, Y., Ni, J., Luo, Q., Chen, J., Zhang, H., Liu, J., Han, C., Fu, K., Ji, C., Cai, X., Hao, J., Zheng, Z., Xu, S., Chen, J., Yu, X., Zhang, Q., Qian, D., Wang, S., Zhou, H.: MICCAI STS 2024 challenge: Semi-supervised instance-level tooth segmentation in panoramic X-ray and CBCT images. Medical Image Analysis **110**, 103986 (2026). https://doi.org/10.1016/j.media.2026.103986
20. Wang, Y., Zhang, Y., Chen, X., Wang, S., Qian, D., Ye, F., Xu, F., Zhang, H., Dan, R., Zhang, Q., Huang, X., Huang, Z., Liu, J., Zheng, Z., Wu, C., Li, Y., Li, Z., Ma, Z., Cui, W., Luo, S., Wang, C., Chen, Y., Li, T., Liu, Y., Feng, X., Ni, J., Liu, D., Wang, Q., Lin, Z., Song, W., Li, Y., Wang, B., Wang, C., Chen, Q., Li, M., Zhou, H., Jin, Q.: MICCAI 2023 STS Challenge: A retrospective study of semi-supervised approaches for teeth segmentation. Pattern Recognition **170**, 112049 (2026). https://doi.org/10.1016/j.patcog.2025.112049
21. Wang, Y., Solomon, J.: Deep Closest Point: Learning Representations for Point Cloud Registration. In: 2019 IEEE/CVF International Conference on Computer Vision (ICCV). pp. 3522–3531 (2019). https://doi.org/10.1109/iccv.2019.00362